Course design of GFN1000

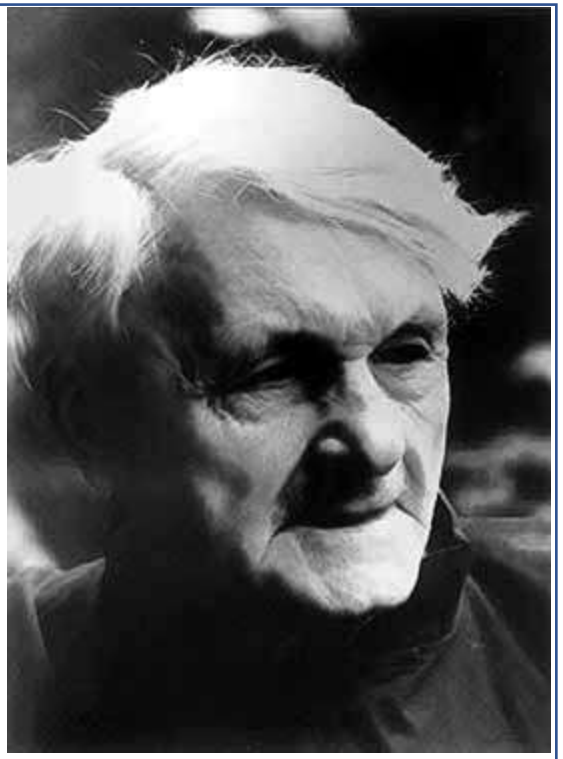
- Part 1. Human exploration of the physical universe
- Part 2. Human exploration of the world of life
- Part 3. Our understanding of human understanding
  - Text 7: Poincaré, *Science and Method*
  - Text 8: Kandel, *In Search of Memory*
  - Text 9: Needham, *The Shorter Science and Civilization in China*
  - Text 10a: Nathan Sivin, ‘Why the Scientific Revolution Did Not Take Place in China – or Didn’t it?’
  - Text 10b: 沈括：《梦溪笔谈》
  - Text 11a: Dunham, *The Mathematical Universe*
  - Text 11b: Euclid, *Elements*

GFN 1000
In Dialogue with Nature

Joseph Needham

Core Question

What do Chinese know about Nature?



Joseph Needham
(1900-1995)

List of sinologists

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- Mark Elliott
- Mark Elvin
- Nathan Sivin
- Jonathan Spence
- Richard B. Stamps
- Hugh Stimson (1931–2011)
- Laurence Thompson
- Tsien Tsuen-hsuin (T.H. Tsien)

Sinology (汉学)

- An academic discipline that focuses on the study of China primarily through Chinese philosophy, language, literature, culture and history.

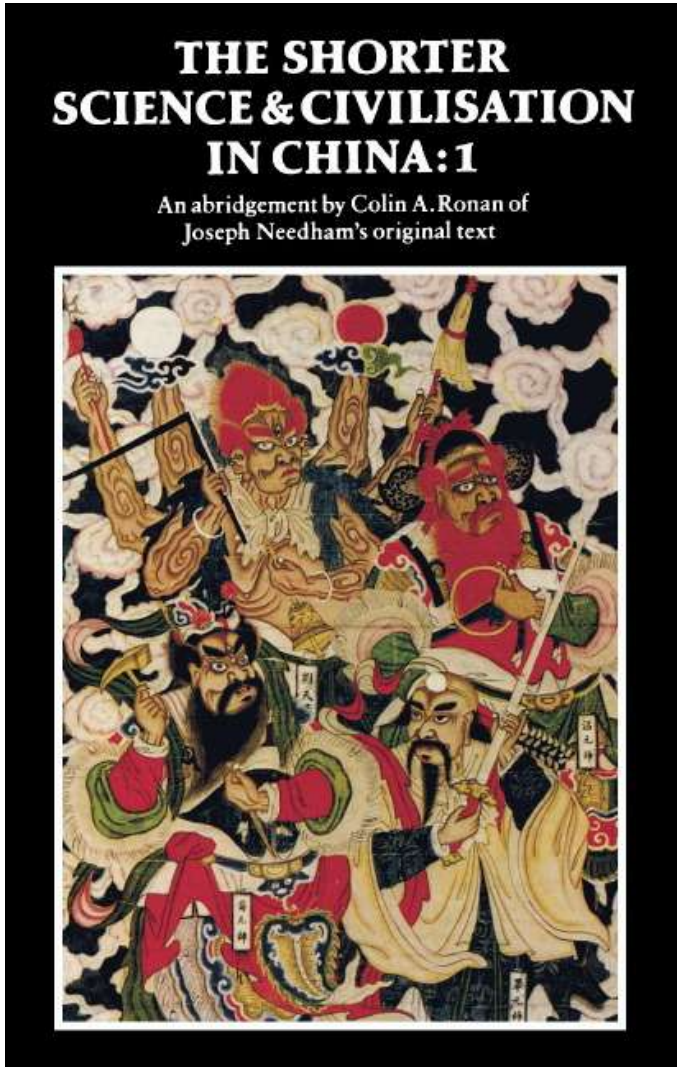
- Now in the Western world:

  More about the study of classical Chinese language and literature.

http://en.wikipedia.org/wiki/Sinology

Text

- Colin Ronan, *The Shorter Science and Civilization in China* Vol. 1.
- An abridged version (5 volume condensation).



Science and Civilization in China

-ongoing series of books about the history of science and technology in China

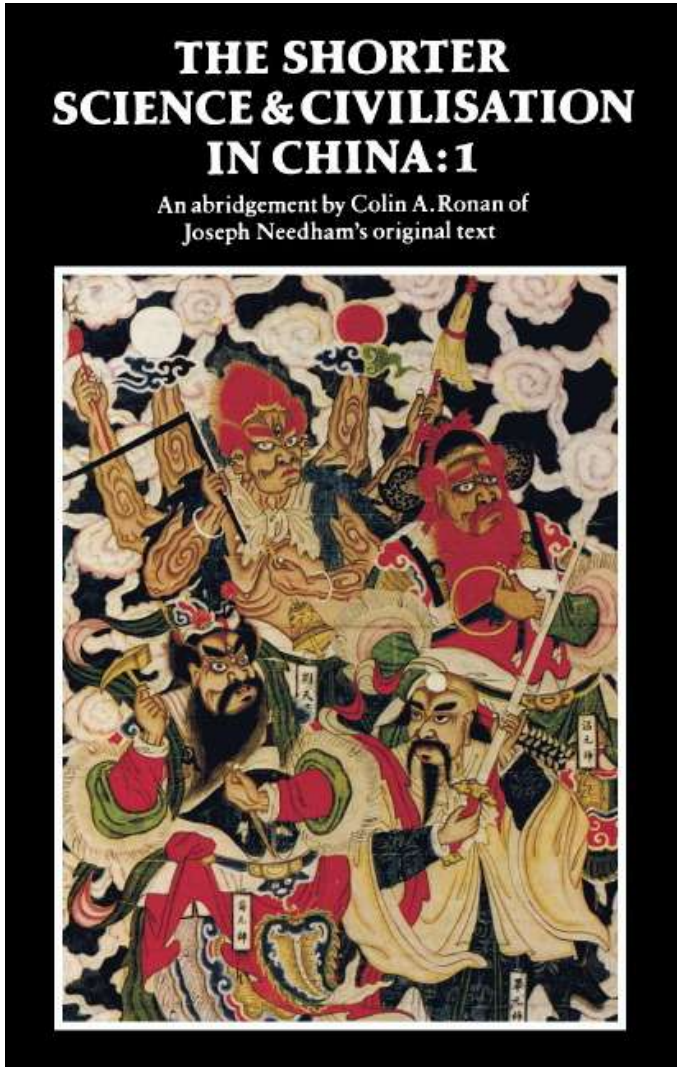


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7 volumes, 27 books published so far, since 1954.

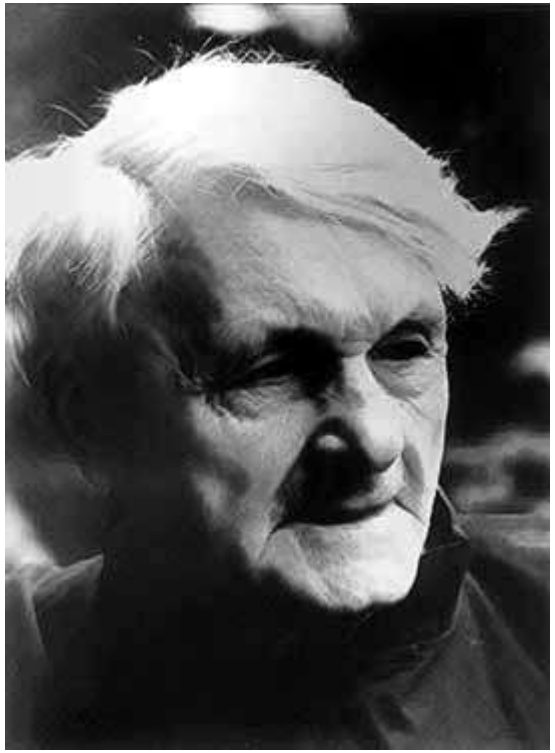
Text

- Colin Ronan, *The Shorter Science and Civilization in China* Vol. 1.
- An abridged version (5 volume condensation).
- Part of Chapter 10.



Joseph Needham (1900-1995)

- Full name: Noel Joseph Terence Montgomery Needham
- Known mainly by his Chinese name: 李约瑟.
- British biochemist, historian and sinologist.
- Received bachelor’s (1921), master’s (Jan 1925) and doctoral (Oct 1925) degrees from Cambridge University.
- Then worked at Cambridge, specializing in embryology (胚胎学) and morphogenesis (发育学).
- Elected as a fellow of the Royal Society in 1941, and a fellow of the British Academy in 1971. In 1992, he was awarded the Companionship of Honor. The Royal Society noted he was the only living person to hold these three titles.
- Best known for his works on the history of Chinese science.
- After Needham's death, Cambridge University established the Needham Research Institute, a center for research into the history of science, technology and medicine in East Asia.



Credit: Needham Research Institute

Joseph Needham (1900-1995)



Joseph Needham in his biochemistry lab in Cambridge, England, 1930s.

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Joseph Needham (1900-1995)

- 1936: Three Chinese scientists, Lu Gwei-djen (鲁桂珍), Wang Ying-lai (王应睐), and Chen Shi-chang (沈诗章) came to Cambridge for graduate study.
- Lu Gwei-djen (1904–91) taught him Classical Chinese and ignited his interest in China's technological and scientific past.
- 1942 to 1946: Under the Royal Society's direction, he was appointed as the director of the Sino-British Science Co-operation Office (中英科学合作馆) in Chongqing.
- In 1945, Needham wrote his first book on the history of Chinese technology, *Chinese Science*.
- In 1948, Needham proposed a project to the Cambridge University Press for a book on Science and Civilization in China.
- He devoted much energy to the history of Chinese science until his retirement in 1990.

"命运使我以一种特殊的方式皈依到中国文化价值和中国文明这方面来。"

——《李约瑟文集》

Needham's Grand Question

(or the Needham Question/Paradox)

Why China had been overtaken by the West in science and technology, despite its earlier successes?

Science:
Modern definition: a systematic enterprise that builds and organizes knowledge in the form of testable explanations and predictions about the world.

Four Foci in the Text

1: Introduction.

2-3: Origins of some of the most important Chinese scientific words. ← 1. Chinese words

[4-12 skipped]

13-20: The stabilised Five-Element theory. Various orders of the five elements and quantitative approach by Mohists.

21-23: The symbolic correlations.

24: Numerology and scientific thinking. } 2. Five elements

25-29: The theory of the Two Fundamental Forces. ← 3. *yin-yang*

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34-35: Order and Pattern = Organism, a kind of mysterious resonance.

36-40: Comparison between Western thought and Chinese thought in terms of causality, the use of numbers and time/space

41-42: Causality is like an orchestra. } 4. Associative thinking

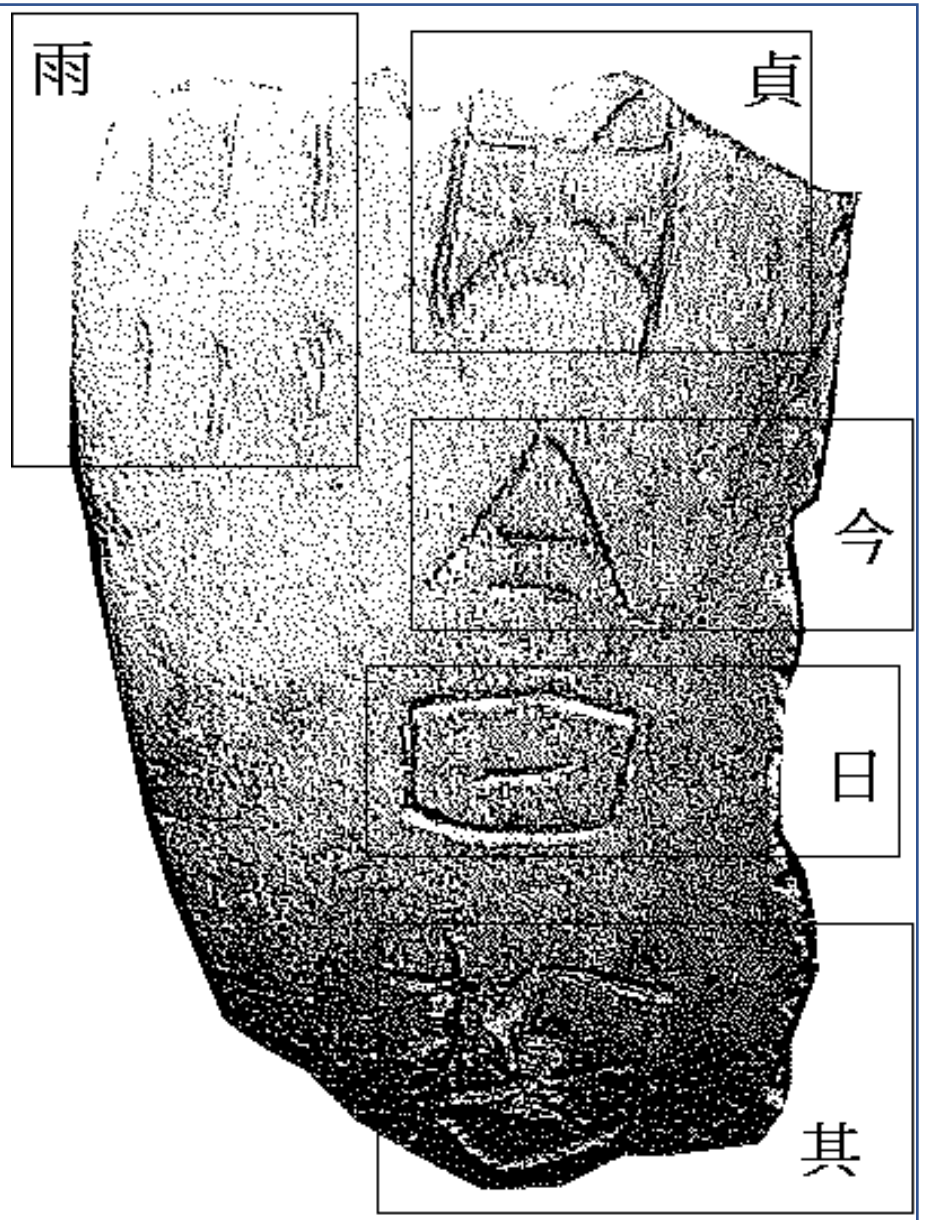
Origins of Some of the Chinese Scientific Words

“ **Before any science can develop there must be a suitable stock of words**, and now, due to the discovery of the **Anyang oracle-bones** already described, as well as the characters inscribed on **Shang and Chou bronze vessels**, there is a copious graphic vocabulary for study. Not all the characters have been identified, but even so there are enough for a selection to be made of those ideographs that throw light on the origins of Chinese scientific terms. It is true that the **ancient words probably had little influence on the thinking of the exponents of the proto-sciences** in Chhin and Han times, but for us these early ideographs are of interest in themselves because **they help us to understand the ancient Chinese approach to science**.”

---From Para 2, Text 9 by Needham

Chinese Words

- Chinese writing first attested around 1300 B.C., with possible earlier precursors.
- Unique local signs and unique principles.
- Evolved independently.
- The oldest preserved Chinese writing of the late Shang Dynasty consists of **religious divination** (占卜) about dynastic affairs.



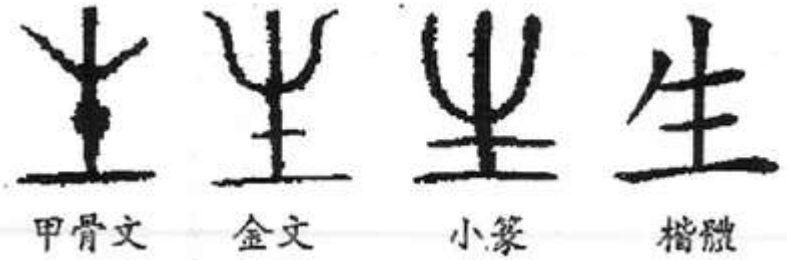
貞：今 日，其 雨？

Test: Today, will it rain?

Chinese Words



Drawing of a ritual vessel containing two pieces of jade (玉).



This character looks like a plant growing out of the ground.

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41-42: Causality is like an orchestra. } 4. Associative thinking

Hundred Schools of Thought (诸子百家)

- During the Spring and Autumn period and the Warring States period (春秋战国时代), a great expansion in culture and philosophy has occurred. This leads to the hundred schools of thought including:

  - Confucianism (儒家)
  - Taoism (道家)
  - Legalism (法家)

  } Humanity?

  - Mohism (墨家)
  - The School of Names or Logicians (名家)
  - The School of Naturalists or Yin-yang (阴阳家)

  } Nature?

Mozi (墨子)

- Mozi (479 – 381 BC), founder of Mohism (墨家).



- *Mo Ching*《墨经》, a book of his thought, covers topics in epistemology (认识论), logic, astronomy, geometry and physics.

- One branch of Mohism became very similar to the school of Names and are grouped as the Mo-Ming schools (墨名家) in our text.

Tsou Yen (鄒衍)

- Tsou Yen (305 – 240 BC), representative thinker of the School of Yin-yang.

- The School of Yin-yang synthesized the concepts of yin-yang (阴阳) and the Five Elements (五行).

- The Mo-Ming schools has a more analytical thinking than the school of Yin-Yang. This branch of thinking is said to be quite similar to the Greek philosophy.

- However, after the Han dynasty, the Mo-Ming schools have gone extinct while the Yin-yang school has integrated itself into the main stream philosophies – Confucianism and Taoism.



Five Elements (五行)

Element	Process
Water	Soaking, dripping, descending (dissolving)
Fire	Heating, burning, ascending
Wood	Accepting form by submitting to cutting and carving instruments
Metal	Accepting form by moulding when in the liquid state, and the capacity of changing this form by re-melting
Earth	Producing edible vegetation

Needham's comments:

〃 The conception of the elements was not so much of five sorts of fundamental matter as of five sorts of fundamental process. Chinese thought again concentrating on **relation** rather than substance. 〃

Five Elements (五行)

The Cosmogonic Order （宇宙起源序）	w F W M E 水 火 木 金 土
The Mutual Production Order （相生序）	W F E M w 木 火 土 金 水
The Mutual Conquest Order （相克序）	W M F w E 木 金 火 水 土
The ‘Modern Order’ （「現代」序）	M W w F E 金 木 水 火 土

- The elements were supposed to have come into being;
  Water: primeval element
- One gives rise to one another;
  Season: Wood for spring…
- One conquers its predecessor, e.g.,
  Water conquers Fire because it can extinguish it
- Obscure significance

Five Elements (五行)

- The Mutual Production Order

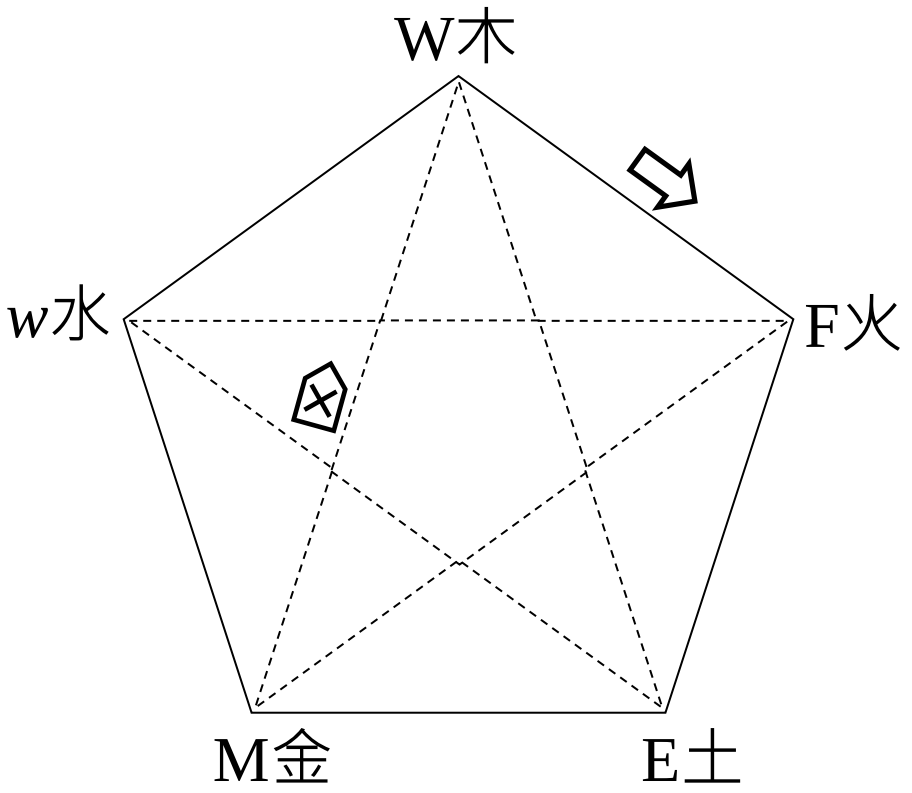
  W ⇨ F ⇨ E ⇨ M ⇨ *w*

  木 ⇨ 火 ⇨ 土 ⇨ 金 ⇨ 水

- The Mutual Conquest Order

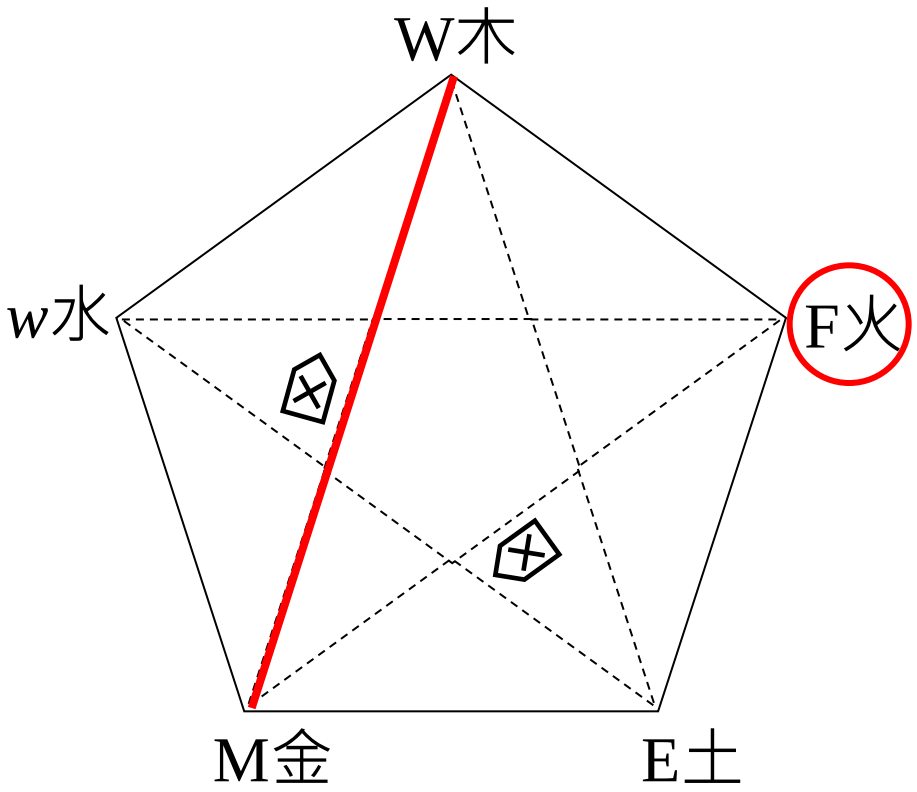
  W ⌫ M ⌫ F ⌫ *w* ⌫ E

  木 ⌫ 金 ⌫ 火 ⌫ 水 ⌫ 土



Principle of Control (相制原理)

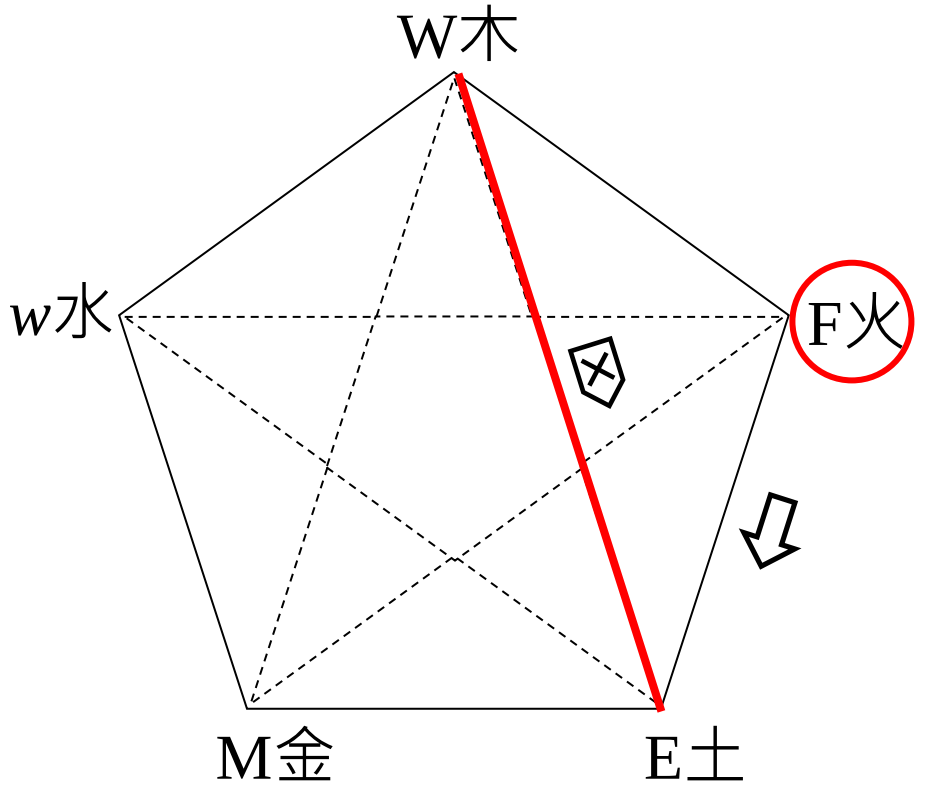
- Derived solely from the Mutual Conquest Order.
- A given process of conquest is controlled by the element that conquers the conqueror.
- Example: F (火) controls the process of M (金) conquering W (木).



“敌人的敌人就是朋友”

Principle of Masking (相化原理)

- Depends on both the mutual production and the mutual conquest orders.
- Mask a process of change by some other process which creates more of the material than is being destroyed or makes it faster.
- Example: W(木) conquers (reduces) E (土) but F(火) masks the process by destroying wood and producing E (土).



A fragment of the criticisms

It is important to note that in both these principles there lurks a strong 20
quantitative element: the conclusions depend on quantities, speeds and rates. This arose, perhaps, from questions raised simply by the Enumeration Orders themselves, for we should realise that the early Chinese thinkers were not satisfied with them. This can be seen, for instance, in the *Mo Ching* where a fragment of the criticisms of the Naturalists by the late Mohists is preserved:

> 'C The Five Elements do *not* perpetually overcome one another.
> CS The five are Metal, Water, Earth, Wood, and Fire. Quite apart [from any cycle] Fire naturally melts Metal, if there is enough Fire. Or Metal may pulverise a burning Fire to cinders, if there is enough Metal. Metal will store Water [but does not produce it]. Fire attaches itself to Wood [but is not produced from it].'*

The Symbolic Correlations

- The five elements gradually came to be associated with every conceivable category of things in the universe that it was possible to classify in fives. [para. 21]

Table 9. ***The symbolic correlations***

Elements *hsing* 行	Seasons *shih* 時	Cardinal points *fang* 方	Tastes *wei* 味	Smells *chhou* 臭	Stems (denary cyclical signs) *kan* 干	Branches (duodenary cyclical signs) and the animals pertaining to them *chih* 支	Numbers *shu* 數
WOOD	spring	east	sour	goatish	*chia i* 甲 乙	*yin* 寅 (tiger) and *mao* 卯 (hare)	8
FIRE	summer	south	bitter	burning	*ping ting* 丙 丁	*wu* 午 (horse) and *ssu* 巳 (serpent)	7
EARTH	—[a]	centre	sweet	fragrant	*wu chi* 戊 己	*hsü* 戌 (dog), *chhou* 丑 (ox), *wei* 未 (sheep) and *chhen* 辰 (dragon)	5
METAL	autumn	west	acrid	rank	*kêng hsin* 庚 辛	*yu* 酉 (cock) and *shen* 申 (monkey)	9
WATER	winter	north	salt	rotten	*jen kuei* 壬 癸	*hai* 亥 (boar) and *tzu* 子 (rat)	6

- It seems that in the beginning these correlations were helpful to scientific thought in China… it was only when they became over-elaborate and fanciful, too far removed from the observation of Nature, that they were positively harmful. [para. 23]

Criticism of the Symbolic Correlations



Wang Chhung (王充)

(27 – c. 100 AD)

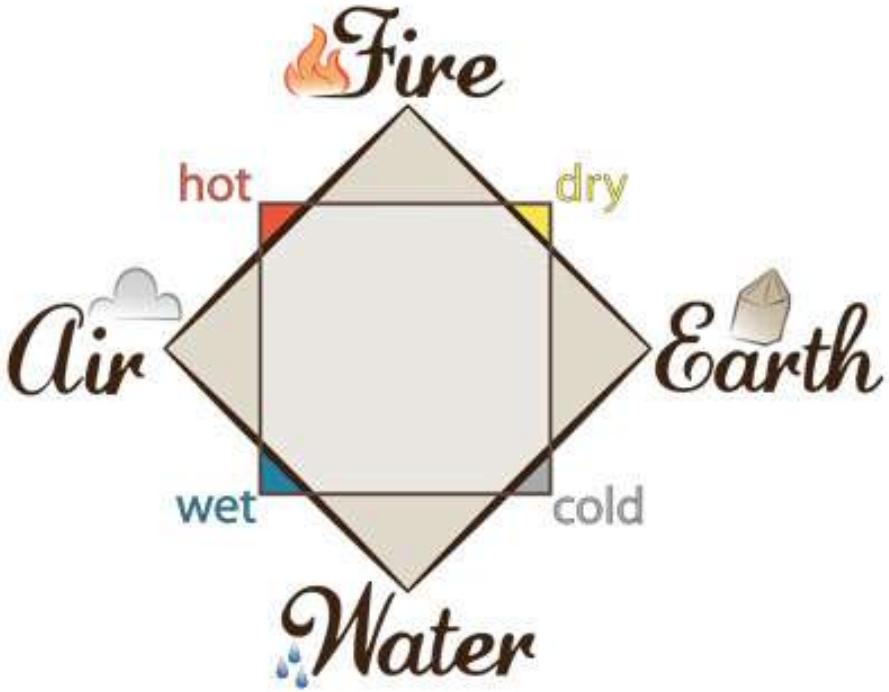
- A Chinese astronomer, meteorologist, naturalist, philosopher, and writer active during the Han Dynasty.
- He developed a rational, naturalistic and mechanistic account of the world and of human beings and gave a materialistic explanation of the origin of the universe.

- He criticised the symbolic correlations of the five elements since it is very often happens that animals do not overpower one another as they ought to do on these theories.

  E.g. 1: Rat is water, horse is fire, then why doesn't the rats attack the horses?

  E.g. 2: Cock is metal, hare is wood, then why doesn't the cocks devour hares?

> 'However when we go into the matter more thoroughly, we find that in fact it very often happens that animals do not overpower one another as they ought to do on these theories. The horse is connected with *wu* (Fire), the rat with *tzu* (Water). If Water really conquers Fire, [it would be much more convincing if] rats normally attacked horses and drove them away. Then the cock is connected with *yu* (Metal) and the hare with *mao* (Wood). If metal really conquers wood, why do cocks not devour hares?'



Aristotle went on to argue that change in form takes place between a pair of opposites or contraries, one of which is the form to be achieved, the other its privation or absence…….Change, for Aristotle, is thus never random, but confined to the narrow corridor connecting pairs of contrary qualities; order is thus discernible even in the midst of change.

Text 2, Para. 14

cf. Humorism (体液学说) in the west

- A system of medicine detailing a supposed makeup and workings of the human body. From Hippocrates onward, the theory was adopted by Ancient Greek and Roman physicians and philosophers.
- Four bodily fluids determined a person's temperament and an imbalance led to certain sicknesses dependent upon which humors were in excess or deficit.

HUMOUR	SEASON	AGES	ELEMENT	ORGAN	QUALITIES	TEMPERAMENT	**Temperament**
Blood	Spring	Infancy	Air	Liver	Moist & Hot	Sanguine	多血质
Yellow Bile	Summer	Youth	Fire	Gallbladder	Hot & Dry	Chloric	胆汁质
Black Bile	Autumn	Adulthood	Earth	Spleen	Dry & Cold	Melancholic	忧郁质
Phlegm	Winter	Old Age	Water	Brain/Lungs	Cold & Moist	Phlegmatic	粘液质

(*Wikipedia*: Humorism)

cf. Bloodletting (放血疗法)

The withdrawal of blood from a patient to prevent or cure illness and disease. The most common medical practice performed by doctors from antiquity up to the late 19th century, a time span of almost 2,000 years.



Bloodletting in 1860

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41-42: Causality is like an orchestra. } 4. Associative thinking

***Chhi* (=*qi*; 气)**

- An active principle forming part of any living thing.
- *Qi* is frequently translated as **life energy, life force, or energy flow**.
- The stuff of which the universe and the celestial bodies are composed.

Two Fundamental Forces

- “One *yin* and one *yang*; that is the *Dao*.” (*I Ching*)
  “一阴一阳之谓道。” 《易经》
- Only two components of *qi* operating in nature, one *yin* and one *yang*.
- Dominating over the other successively in a wave-like action.
  (Rotating the diagram)
- *Yin* and *yang* are both opposite and complementary to each other.



Taijitu liangyi diagram
太极两仪图

Two Fundamental Forces

- Harmony between Yin *chhi* (=*qi*; 气) and Yang *chhi* of Heaven/Earth and man.
- A quote by Tung Chung-Shu 董仲舒《春秋繁露》〈同类相动篇〉:

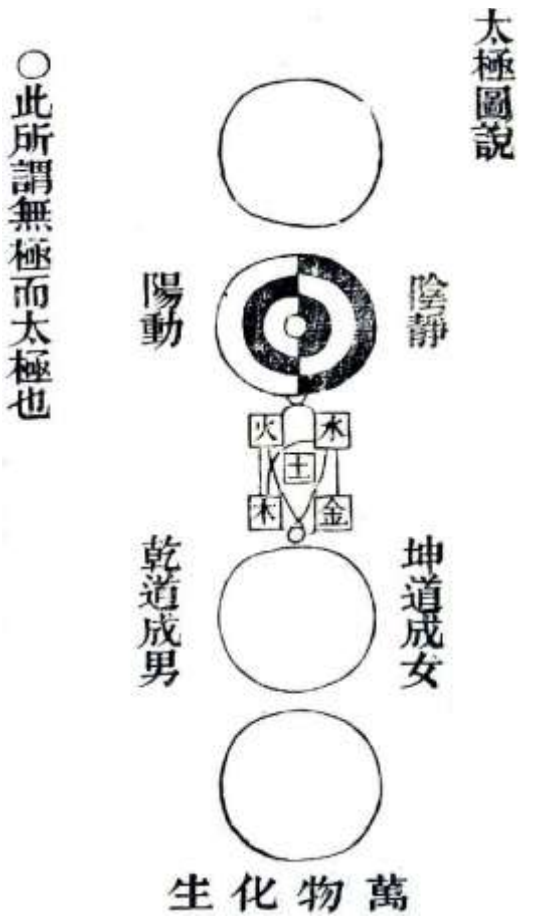
'Heaven has Yin and Yang, so has man. When the Yin *chhi* of Heaven and Earth begins [to dominate], the Yin *chhi* of man responds by taking the lead also. Or if the Yin *chhi* of man begins to advance, the Yin *chhi* of Heaven and Earth must by rights respond to it by rising also. Their Tao is one. Those who are clear about this [know that] if the rain is to come, then the Yin must be activated and its influence set to work. If the rain is to stop, then the Yang must be activated and its influence set to work.'

天有阴阳，人亦有阴阳。天地之阴气起，而人之阴气应之而起，人之阴气起，而天地之阴气亦宜应之而起，其道一也。

(Para. 28)

《太极图说》 ---周敦颐 (宋)

- A Chinese cosmologist, philosopher, and writer during the Song Dynasty.
- He conceptualized the Neo-Confucian cosmology of the day, explaining the relationship between human conduct and universal forces.
- He emphasizes that humans can master their *qi* ("vital life energy") in order to accord with nature.



其宇宙观，“无极而生太极，太极而生阳，动极而静。静极复动，一动一静，互为其根；分阴分阳，两仪立焉。”

Principles of scientific ideas

- Scientific ideas of the Chinese involved two fundamental principles, two forces and five elements.
  - Two forces: yin and yang
  - Five elements: M, W, w, F, E

this practical emphasis of magic, the symbolic correlations of the Chinese were just what a magician would need to practise his art: they brought order into things and did this at a time when no one could know what would bring success and what would not in any magical or experimental procedure. There had to be some guide to choosing the right conditions, and the correlations provided it. If one were experimenting or doing magic with water then it seemed logical, say, not to wear red, the colour of fire. Such associations might be more intuitive than anything else, but at the time what else could they be?

33 Modern scholars have called this kind of mental approach 'associative thinking' or 'co-ordinative thinking'. It is a system that works by association and

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30-42: 'Associative' thinking and its significance.

30-33: the two laws — law of similarity and law of contagion.

34-35: Order and Pattern = Organism, a kind of mysterious resonance.

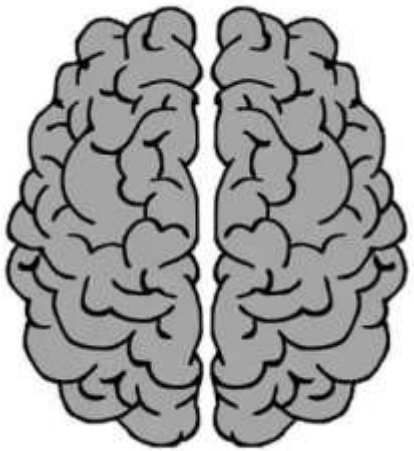
36-40: Comparison between Western thought and Chinese thought in terms of causality, the use of numbers and time/space

41-42: Causality is like an orchestra. } 4. Associative thinking

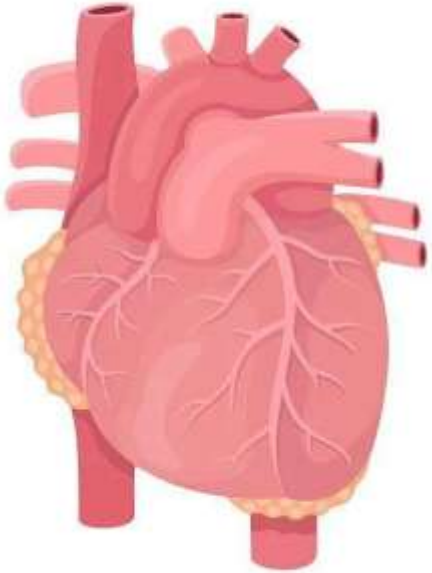
4. Associative Thinking

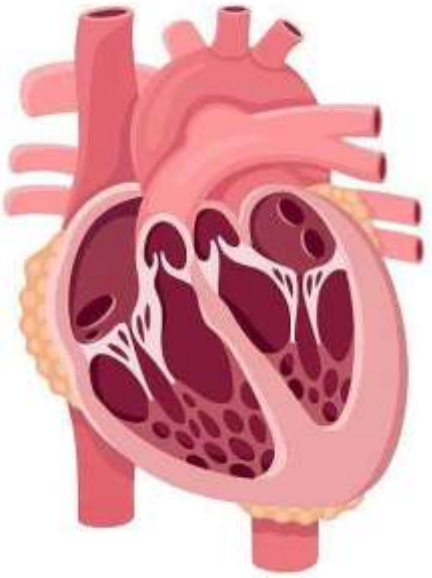
- Associative thinking: A system works by association and intuition, and it has its own logic and its own laws of cause and effect.
  - ➢Law of similarity: according to which like produces like
  - ➢Law of contagion: things were in contact and continue to act upon one another

Law of Similarity (类似律)









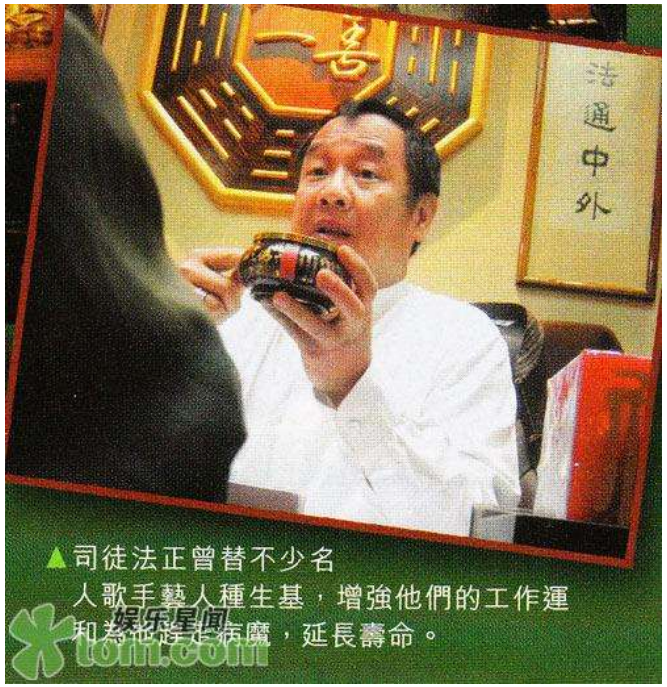
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4. Associative Thinking

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Law of Contagion (接触律)

- *e.g.*「种生基」(creating a live grave生坟)
- A taoism *feng shui* 风水 technique
- Buried the clothes and hair of a living person into a live grave. The person is believed to benefit from the good *yin* dwelling during his/her life time.





4. Associative Thinking

- Associative thinking: A system works by association and intuition, and it has its own logic and its own laws of cause and effect.
  - ➢Law of similarity: according to which like produces like
  - ➢Law of contagion: things were in contact and continue to act upon one another

  Eg. You eat a lot as a pig does, you will become piggish fat.
- cf. Analytical thinking:
  - You eat a lot. You will gain weight because of the nutrient-rich diet.

Characteristics of Chinese thought

- Order
- Pattern

  OR

- Organism

  Existence of things depend on the whole world-organism, and they reacted on one another by a kind of mysterious resonance.

Things of the Same Genus Energize Each Other
---Tung Chung-Shu (董仲舒)

- A Chinese philosopher, politician, and writer of the *Han Dynasty*.
- He is traditionally associated with the promotion of Confucianism as the official ideology of the Chinese imperial state.

'If water is poured on level ground it will avoid the parts which are dry and move towards those that are wet. If [two] identical pieces of firewood are exposed to the fire, the latter will avoid the damp and ignite the dry one. All things reject what is different [to themselves] and follow what is akin. Thus it is that if [two] *chhi* are similar, they will coalesce; if notes correspond they resonate. [The experimental proof of this is extraordinarily clear. Try tuning musical instruments.] The *kung* note or the *shang* note struck upon one lute will be answered by the *kung* or the *shang* notes from other stringed instruments. They sound by themselves. There is nothing miraculous, but the Five Notes being in relation; they are what they are according to the Numbers [whereby the world is constructed].

'[Similarly] lovely things summon others among the class of lovely things; repulsive things summon others among the class of repulsive things. This arises from the complementary way in which a thing of the same class responds—as for instance if a horse whinnies another horse whinnies in answer, and if a cow lows, another cow lows in response.

'When a great ruler is about to arise auspicious omens first appear; when a ruler is about to be destroyed, there are baleful ones beforehand. Things indeed summon each other, like to like, a dragon bringing rain, a fan driving away heat. .

Needham’s Grand Question

(or the Needham Question/Paradox)

Why China had been overtaken by the West in science and technology, despite its earlier successes?

Why did Modern Science not Emerge in China?

Needham's answer

- "In sum, I believe that the analyzable differences in social and economic pattern between China and Western Europe will in the end illuminate… both the earlier predominance of Chinese science and technology and also the later rise of modern science in Europe alone." (SCC7:2 p. 23)

- Needham suggested it was the rise of the bourgeoisie (城市资产阶层) (rich people who owned property) in Western Europe from 15th century that gave rise to science.

- Reading Assignment: text 9.